Indian Institute of Engineering Science and Technology, Shibpur

B.Tech 4^{th} Semester End-Semester Examination,

May 2026.

Design and Analysis of Algorithms (CS2201)

Full Marks: 50

Time: 3 Hours

Answer any five(5) questions. It is advised to answer all parts of a question together. Credit will be given to precise answer.

1. (a) What are the different problem solving strategies you know of? Explain them briefly with suitable Example?

(b) What do you mean by maximal independent set in a graph $G = \{V, E\}$. Propose an algorithm to find the maximal independent set in a tree.

(c) Prove that all leaves must be there in the result of your proposed algorithm above. [4+3+3]

2. (a) You are given a set of points along a line. Propose an efficient algorithm to find the closest pair among them and hence give an estimation of running time of your algorithm.

(b) Extend the algorithm that you proposed above for set of points in 2-dimension and also provide an estimate of its running time. [4+6]

3. (a) Explain Dijkstra's algorithm to find shortest path between a source (s) and the target (t) vertex in an weighted undirected graph.

(b) Explain the data structure used to implement Prim's algorithm for minimum spanning tree with proper justification. [6+4]

4. (a) Briefly explain the Kruskal algorithm to find the minimum spanning tree of an un-directed weighted graph and provide the proof of correctness of the algorithm.

(b) What do you mean by 'union by rank' and 'union by height'? Prove that in both cases $find()$ operation can be performed in $n \lg n$ time. [(3+3)+4]

5. (a) Define the class NP and hence prove that SAT belongs to NP.

(b) State Cook's theorem and its key aspect.

(c) Transform the following search version of the Clique problem into its equivalent decision version and hence prove that they both (search and decision version) are equally hard.

Problem: Clique (search) Input : A graph $G = \{V, E\}$ Output: find a clique of largest size

[2+3+5]

6. (a) Suppose you are given following two problems

Problem: Maximum Matching Input : A graph $G = \{V, E\}$ Output: Matching (edges) of Maximum Size	Problem: Perfect Matching Input : A graph $G = \{V, E\}$ Question: Does G have Perfect Matching?

Assuming there is an efficient algorithm for 'Perfect Matching', design an efficient algorithm for 'Maximum Matching'.

(b) Prove that vertex cover problem as given below is NP-Complete.

Problem: Vertex Cover
Input : A graph G and a +ve integer K
Question: Does G contain a vertex cover of size at most K?

[6+4]

7. (a) What are the properties a problem must posses to be solved by dynamic programming?
   (b) Given a chain of four matrices A_1, A_2, A_3, and A_4 with dimensions $5 \times 4, 4 \times 6, 6 \times 2$ and 2×7 respectively. Find the optimal sequence of multiplication i.e. $m(1, 4)$.

[4+6]